

Interface Control Guide

PacketXpress Version 5.1.0

AXELLIO

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1 Introduction

Axellio® PacketXpress® is a suite of tools providing the ability to manage fast moving network packet data at speeds, capacities and efficiencies previously not feasible. PacketXpress is built on an patented “clean sheet” special purpose design architecture leveraging modern industry standard COTS HW capabilities that allows network data rates at 200Gbps+ to be captured, stored and distributed to radically simplify and lower the solutions cost of high-speed streaming data use cases.

To enable interaction with external applications, PacketXpress provides a RESTful web service application programming interface (*axapi*). Applications communicating with PacketXpress using the *axapi* interface can access and manage network capture data over a standard HTTPS communications channel.

The axapi interface is provided as a stable and fully tested mechanism for accessing the PacketXpress system.

2 The axapi RESTful API

2.1 Security

API security is provided through two layers of protection:

- SSL/TLS v1.2 provided by HTTPS
- JWT token-based authentication
    - Every request requires a JWT be attached in the authorization header of the request. Conforming with JWT standards the authorization header must start with “Bearer” followed by a single space, followed by the token string.
    - This token contains information about the user at the time it was generated. This user information is used to determine login status and permissions.
    - Any request that has a missing, bad, or expired token will receive the following payload on the response:

URI	api/resource/<any_resource>	

Response	{
	“code”: 401,
	“message”: “Unauthorized”,
	“errors”: [{
	“domain”: “Authorization”,
	“reason”: string,
	“message”: string,
	“instance”: Error | NULL,
	}]
	}
		
		HTTP Status code
		HTTP Status string
		Array of errors that happened
		Where the error occurred
		Name of the error
		Why the error occurred
		Instance of the error object

Response Status Code	401	Unauthorized

2.2 Interface Definition By Functional Area

2.2.1 Security

2.2.1.1 API Login (Get Token)

URI	/axapi/login	Request a login token using username and password
Method	POST	
Headers	Content-Type: application/json	
Payload (JSON)	{	

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	“username”: string, “password”: string }	[required] Username for the user [required] Password for the user

Response	{
	"token": string,
	}
	
	{
	“message”: string,
	}
		Response for 200 status code
		The newly generated token
		
		Response for 400/401 status code
		The reason for the error

Response Status Codes	201 400 401	OK Bad request, something invalid/missing in the payload Username or password does not match

2.2.1.2 API Logout

URI	/axapi/logout	Request a logout
Method	POST	
Headers	Content-Type: application/json Authorization: “Bearer <token>”	

Payload (JSON)	{
	“token”: string,
	}
		The token to be logged out

Response	{
	"token": string,
	}
	
	{
	“message”: string,
	}
		Response for 200 status code
		The newly generated token
		
		Response for 400/401 status code
		The reason for the error

Response Status Codes	200 400 401	OK Bad request, something invalid/missing in the payload Missing/invalid token provided

2.2.2 Data Extract

URI	/axapi/pcap	Returns a PCAP file
Method	GET	
Headers	authorization: "Bearer <token>"	
Payload (query string)	start_time=<time>& end_time=<time>& filter=<filter>& stream_ids=<stream_ids>& archive=<archive_ip> & servers=<server_list>	Optional sub-second epoch time for end of extract Optional sub-second epoch time for end of extract Optional Berkeley Packet Filter Optional stream IDs to extract (default “0-11”) Optional address of archive system Optional use multi-extract; list of cluster systems, comma-separated, is optional
Response	Raw Data	Response for 200 status code
Response Status Codes	200 401 418 500	OK, file transfer Missing/invalid token provided Extraction did not provide any data (no output available) An error occurred while extracting

3 Examples

This python3 script shows how an application can connect to the axapi interface, login, and download a PCAP file, and validates that logout successfully revokes access to session.

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```
#!/bin/python3
import urllib3
import requests
import json
import sys
import subprocess
urllib3.disable_warnings()

server="dev06.swlab.axellio.dom" # Server running Axellio Xpress
Platform
username="fxadmin"               # Username
password="password"              # Password
starttime="1728484411.293105240" # Start time, use pxls -aR --subsec
endtime="1728484411.295105240"   # End time, use pxls -aR --subsec
bpfFilter=""                     # Blank for all packets or use a BPF
Filter (see man 7 pcap-filter) or Tuple Filter
(source_ip,dest_ip,source_port,dest_port,protocol)
streamIds="0"                    # A list of stream IDs to extract data
from. The list may be specified as numbers and ranges using "-" or ".."
with comma separators between entries.
                                 # "n[,n][,n-m][,n..m]" where n and m
are integer values in the range 0-4294967296
filename="test.pcap"             # Name of the extract pcap file.  May
include the fully qualified path.

# Set up the session
s = requests.Session()

# URL for login, make the post request and save the token
print("\nLogging in and getting a session token")
url = "https://%s/axapi/login" % server
r = s.post(url, json={"username":username, "password":password},
verify=False)
r.cookies.get_dict()
token=json.loads(r.text)["token"]
print("Session Token: %s" % token)

# Form the pcap extract URL, start the download and save to filename
print("\nAttempting a pcap extract")
download =
"https://%s/axapi/pcap?filter=%s&start_time=%s&end_time=%s&stream_ids=%
s" % (server, bpfFilter, starttime, endtime, streamIds)
t = s.get(url = download, verify=False,stream=True)

try:
```

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```
    t.raise_for_status()  # Raise an exception for bad status codes
    with open(filename, 'wb') as file:
        for chunk in t.iter_content(chunk_size=8192):
            file.write(chunk)
except Exception as e:
    print("[ERROR] File download failed: %s" % e)
    sys.exit(1)

result = subprocess.run(["ls", "-al", filename],
stdout=subprocess.PIPE, text=True)
print(result.stdout)

print("\nFile %s has been downloaded" % filename)

print("\n")

# Logout and nullify the token
print("Trying to logout")
url = "https://%s/axapi/logout" % server
r = s.post(url, json={"username":username, "password":password,
"token": token}, verify=False)
print("Logged out status: %s" % r)

# Error test to attempt a pcap extract without a valid token
print("Attempting to download again without authorization")
download =
"https://%s/axapi/pcap?filter=%s&start_time=%s&end_time=%s&stream_ids=%
s" % (server, bpfFilter, starttime, endtime, streamIds)
t = s.get(url = download, verify=False,stream=True)

try:
    t.raise_for_status()  # Raise an exception for bad status codes
    with open(filename, 'wb') as file:
        for chunk in t.iter_content(chunk_size=8192):
            file.write(chunk)
except Exception as e:
    print("[ERROR] File download failed4: %s" % e)
    sys.exit(1)

print("axapi test has completed")
sys.exit(0)
```

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4 Support

4.1 Customer Support

Support Portal: axellio.com/support

Email: support@axellio.com

Phone: (800) 463-0297

4.2 Contact Us

Website: axellio.com

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Address: 2375 Telstar Dr. Suite 150, Colorado Springs, CO 80920

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